

SMARTAssist Hub

Version 7.0 — System Documentation

A Multi-Agent AI Assistant Platform for District Education Offices (PPD)
and the Ministry of Education Malaysia (KPM)

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1. Introduction

SMARTAssist Hub is a web-based application that uses Artificial Intelligence (AI) to help education officers carry out their daily administrative work faster and more accurately. Instead of one general chatbot, the system provides **five specialised AI assistants** (called *agents*), each trained for a specific type of task — analysing data, writing official letters, generating reports, proofreading documents, and answering questions about KPM systems and policies.

Think of it as a team of five digital colleagues. When you open the application, you choose which colleague you need, and that colleague greets you, asks the right questions, and produces the output in the official KPM format — complete with letterheads, logos, and the required phrases such as “MALAYSIA MADANI” and “BERKHIDMAT UNTUK NEGARA”.

All conversations are held in Bahasa Malaysia (with an English interface option), and the final documents can be downloaded as Microsoft Word files, PowerPoint slides, or PDF, or sent directly by e-mail.

2. Who Is the System For?

- **PPD / district education officers** — who prepare letters, memos, one-page reports and data summaries every week.
- **School administrators** — who need quick, correctly-formatted official correspondence.
- **Teachers and staff** — who need help using KPM systems such as EMIS, APDM, DTPCare and SK@S.

No technical knowledge is required. The user simply types in plain language — for example “Tulis surat jemputan bengkel ICT” — and the system guides them step by step.

3. The Five AI Agents

Data Analysis (Analisis Data)	Reads uploaded CSV or Excel files and answers questions about the data. It produces summaries, statistics, interactive charts and tables, and always warns the user about data limitations (e.g. small sample size). Results can be exported to PowerPoint or PDF.
Official Letter Generator (Penjana Surat Rasmi)	Interviews the user one question at a time (reference number, date, recipient, subject, etc.), writes the letter body automatically, and assembles a complete official letter or internal memo in KPM format. The finished letter can be downloaded as a Word file with the official letterhead, or e-mailed directly.
Report Generator (Penjana Laporan)	Builds a One Page Report (laporan satu muka surat) for programmes and activities. It collects the programme details, then writes the summary (rumusan) and recommendations (cadangan) automatically in formal government style.

Document Reviewer (Semakan Dokumen)	Acts as a careful editor. The user uploads a PDF or Word file (or pastes text), and the agent checks grammar, spelling, tone, KPM formatting rules and unfilled placeholders. It gives a grade (A–D) and a list of issues, but only rewrites the document if asked.
KPM Support (Sokongan KPM)	A friendly helpdesk assistant for KPM systems (EMIS, APDM, DTPCare, SK@S) and policies. It searches an internal library of official KPM documents and bases its answers on those documents, always citing the source. It replies in plain, WhatsApp-style text.

4. How the System Works (In Simple Terms)

When a user types a message, the following happens behind the scenes, usually within a few seconds:

- **Step 1 — The web page sends the message** to the server (the “brain” of the system running in the background).
- **Step 2 — The server passes the message to the chosen agent.** Each agent has its own detailed set of instructions (called a *system prompt*) that tells the AI how to behave — its personality, its rules, and the exact output format required.
- **Step 3 — The agent consults the Large Language Model (LLM)**, the AI engine that understands and produces language. The agent sends the conversation history plus its instructions, and receives a draft answer.
- **Step 4 — The server checks the answer.** For agents that produce structured output (charts, letters, reports), the system verifies the format is valid. If not, it automatically asks the AI to try again — the user never sees the failure.
- **Step 5 — The web page displays the result** — as a chat bubble, an interactive chart, a document preview, or a review checklist, depending on the agent.

Each conversation is a **session**. Sessions are saved so the user can close a chat, come back later, and continue where they left off. Every agent also greets the user when a session starts and thanks them when it ends.

5. The AI Engine (Large Language Model)

The intelligence behind every agent is a **Large Language Model (LLM)** — the same category of technology behind ChatGPT. SMARTAssist Hub connects to the **DeepSeek** LLM through an internet API (a secure machine-to-machine connection).

A useful way to understand the division of work:

- The **LLM** is like a very knowledgeable, very fast writer who can produce text on any topic.
- The **agents** are like job descriptions handed to that writer: “Today you are a government letter specialist. Follow the KPM format exactly. Ask one question at a time. Never invent personal details.”
- The **server** is the supervisor who checks the writer’s work before it reaches the user.

Important safeguards built into the prompts: agents must not invent data or personal information, must ask when unsure, must base KPM Support answers on real reference documents, and must flag any unfilled placeholder before a document is approved.

6. Technologies and Tools Used

Web server	Python + FastAPI	The “brain” that receives requests, runs the agents, and returns answers.
AI engine	DeepSeek LLM (via API)	Understands and writes Bahasa Malaysia / English text.

User interface	HTML, CSS, JavaScript	The pages, buttons, chat bubbles and animations the user sees.
Charts	Chart.js	Draws the interactive bar, line and pie charts.
Data handling	pandas	Reads and summarises the uploaded CSV / Excel files.
Word documents	python-docx	Builds the downloadable .docx letters and reports, including letterheads.
PDF & slides	reportlab, python-pptx	Exports analysis results to PDF and PowerPoint.
PDF reading	pdfplumber	Extracts text from uploaded PDF files for the Document Reviewer.
Document search	TF-IDF text search	Finds the most relevant KPM reference documents for the Support agent.
E-mail	SMTP	Sends finished documents directly to a recipient's inbox.

7. Key Features

7.1 Guided Document Creation

The letter and report agents never overwhelm the user with a long form. They ask for one piece of information at a time, keep track of what has already been provided, show which fields are still missing, and write the content sections (letter body, report summary, recommendations) automatically. The finished document appears in an editable preview and in a Word-like full-page view.

7.2 Letterhead and Logo Management

Administrators can upload the office letterhead (for letters) and logo (for reports) once, through a dedicated settings page. Every document generated afterwards automatically carries the correct image — on screen and in the downloaded Word file.

7.3 Quick Document Updates

After a document is complete, the user can simply say “kemaskini tarikh kepada 15 Julai” (update the date to 15 July). The system recognises update keywords and edits only the requested part, instead of restarting the whole interview.

7.4 File Upload and Analysis

CSV and Excel files are read instantly, summarised (rows, columns, missing values, statistics), and kept safely on disk so the analysis survives even if the server restarts. PDF and Word files uploaded to the Document Reviewer are converted to text automatically.

7.5 User Feedback and Admin Dashboard

Every AI answer carries thumbs-up / thumbs-down buttons. An administrator dashboard shows total sessions and messages, usage per agent, positive/negative feedback per agent, and the most recent sessions — useful for measuring adoption and answer quality.

7.6 Bilingual Interface and Session History

The interface can switch between Bahasa Malaysia and English with one click. A history panel lists past sessions per agent, and any session can be reopened or deleted.

8. Built-In Quality and Reliability Measures

- **Automatic retry** — if the AI returns an answer in the wrong format, the system silently asks it to correct itself before showing anything to the user.
- **Placeholder detection** — a document cannot be approved for download while any “[PLACEHOLDER]” text remains unfilled.
- **Grounded answers** — the KPM Support agent answers from a library of real KPM documents and names its sources, reducing the risk of made-up information.
- **Data honesty rules** — the Data Analysis agent may only use numbers that actually exist in the uploaded file, and must state caveats when the data is weak.
- **Graceful errors** — if the AI service is temporarily unreachable, the user receives a polite message (and, for KPM Support, a list of relevant documents to read manually) instead of a crash.

9. Current Limitations and Planned Improvements

The system is fully functional, but the following areas are known gaps that are planned for future versions:

- **User login and access control** — at present anyone with the address can use the system. Authentication (for example a staff login) is the top priority before wide deployment.
- **Permanent database** — chat history and statistics are currently kept in the server's memory; a proper database will make them survive restarts. (Uploaded data files are already stored on disk.)
- **Dependence on the AI service** — if the external LLM service is down, the agents cannot generate new answers.
- **Streaming responses** — long answers currently appear all at once; showing them word-by-word would feel faster.

10. Summary

SMARTAssist Hub V.7 brings together five specialised AI assistants in a single, easy-to-use web application tailored to the real paperwork of Malaysian district education offices. It combines a modern chat interface, strict KPM document formatting, automatic content writing, data analysis with charts, document proofreading, and a knowledge-based helpdesk — while keeping the user in control through previews, edits and feedback at every step. With the addition of user login and a permanent database, the system will be ready for full-scale deployment across schools and district offices.